

# JACOB J. POTTS

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## EDUCATION

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**Blended B.S./M.S. in Computer Engineering**  
*Four-year accelerated program leading to both degrees.*  
**California State University, Fullerton**

**Aug 2024 - May 2028**  
**GPA: 3.32**

**Relevant Coursework:** Circuit Analysis | C++ Object-Oriented Programming | Digital Logic | Computer Organization & Assembly | Calculus I-III | Linear Algebra & Differential Equations | Modeling & Simulation of Signals & Systems

## EXPERIENCE

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**Embedded Software Engineering Intern | RAVE Aerospace - Brea, CA**

**Jun 2026 - Aug 2026**

- Developed automated hardware acceptance and production test software that detected USB-C power configurations, automatically applied hardware-specific power requirements, and alerted users to insufficient power, contributing to successful qualification testing at IEE, an independent testing laboratory.
- Validated test-machine networking with bidirectional iPerf testing and supported a production burn-in testing rack by configuring its switch, diagnosing USB-hub related power shutdowns, identifying 7 of 10 faulty cables, and correcting cabling issues to improve system reliability.
- Added real-time Bluetooth connection status to a qualification-testing GUI, with automatic hardware compatibility detection and responsive 1080p/4K scaling, providing immediate operator feedback and connection troubleshooting during testing.
- Developed hardware abstraction layer (HAL) functionality for an embedded Linux system, creating a consistent interface for retrieving device identification, hardware revisions, temperatures, voltages, watchdog status, and GPIO data for automated testing and diagnostics.
- Added automatic hardware identification to performance verification test software and resolved a boot-state failure by tracking connected hardware and updating DHCP configuration files, preventing outdated configurations from being applied when switching between devices.

**General Merchandise Clerk | Stater Bros. Markets - Corona, CA**

**Sep 2024 - May 2026**

- Balanced a 24-hour weekly work schedule with five days of in-person university coursework while supporting customers, maintaining inventory, and collaborating in a fast-paced team environment.

## TECHNICAL SKILLS

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**Programming:** C, C++, Bash, Python, VHDL, MATLAB, Verse, QML

**Frameworks & Tools:** Git, GitHub, SVN, Bitbucket, Linux, VS Code, Minicom, Questa, Multisim, Qt

**Hardware:** GPIO, EEPROM, Network Switch, USB-C Power Testing, Oscilloscope, Multimeters, Function Generator, DC Power Supply, Breadboards

## ENGINEERING PROJECTS & COURSEWORK //TO UPDATE with school and personal projects

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- Circuit & Digital Logic: Built/tested resistor circuits; designed combinational/sequential logic; simulated VHDL in Questa and circuits in Multisim.
- Computer Organization & Assembly: Programmed with registers, memory addressing and bitwise operations; analyzed low-level hardware/software behavior.
- CAD & Design: Designed dimensioned parts and mechanical components using AutoCAD and Fusion 360.